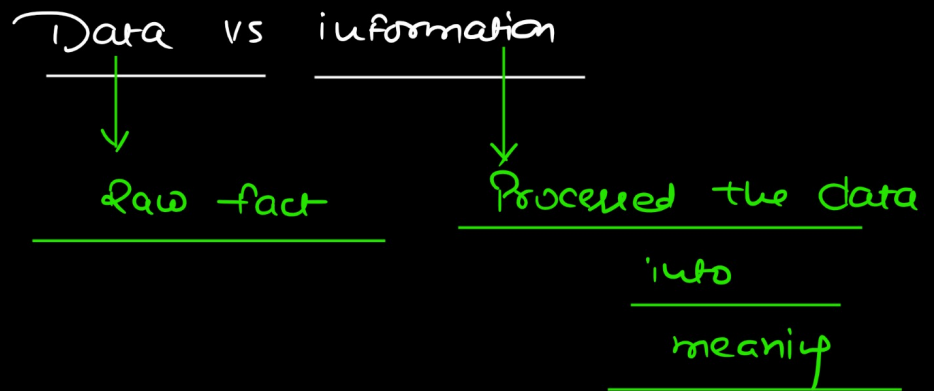
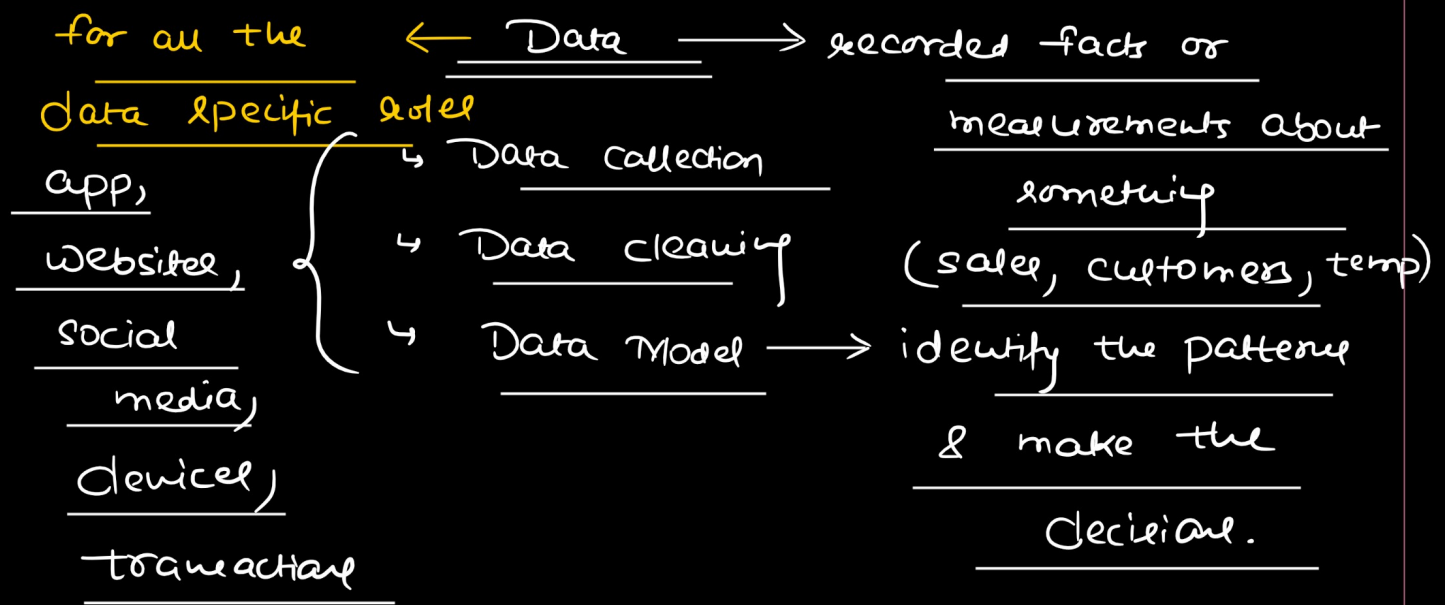




Type of Data

- ① Structured data → rows/columns
order table
transaction data
- ② semi-structured data → logs
from the website, JSON
- ③ unstructured data → images, videos, product reviews,

④

Multimodal data → diff

A single Instagram post may include:

1. image (visual data)
2. caption (text)
3. hashtags (categorical/text)
4. location tag (structured data)
5. background music (audio)

type 4

data in
one file

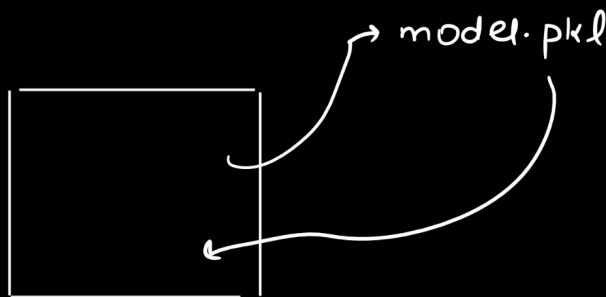
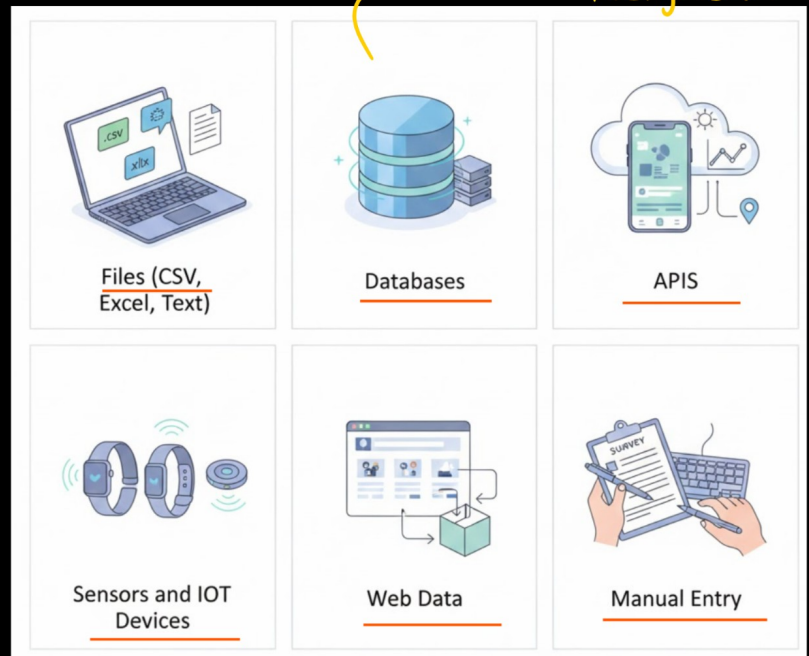
SQL

RDBMS → MySQL,
PostgreSQL

Data Acquisition

common ways to

collect the
data

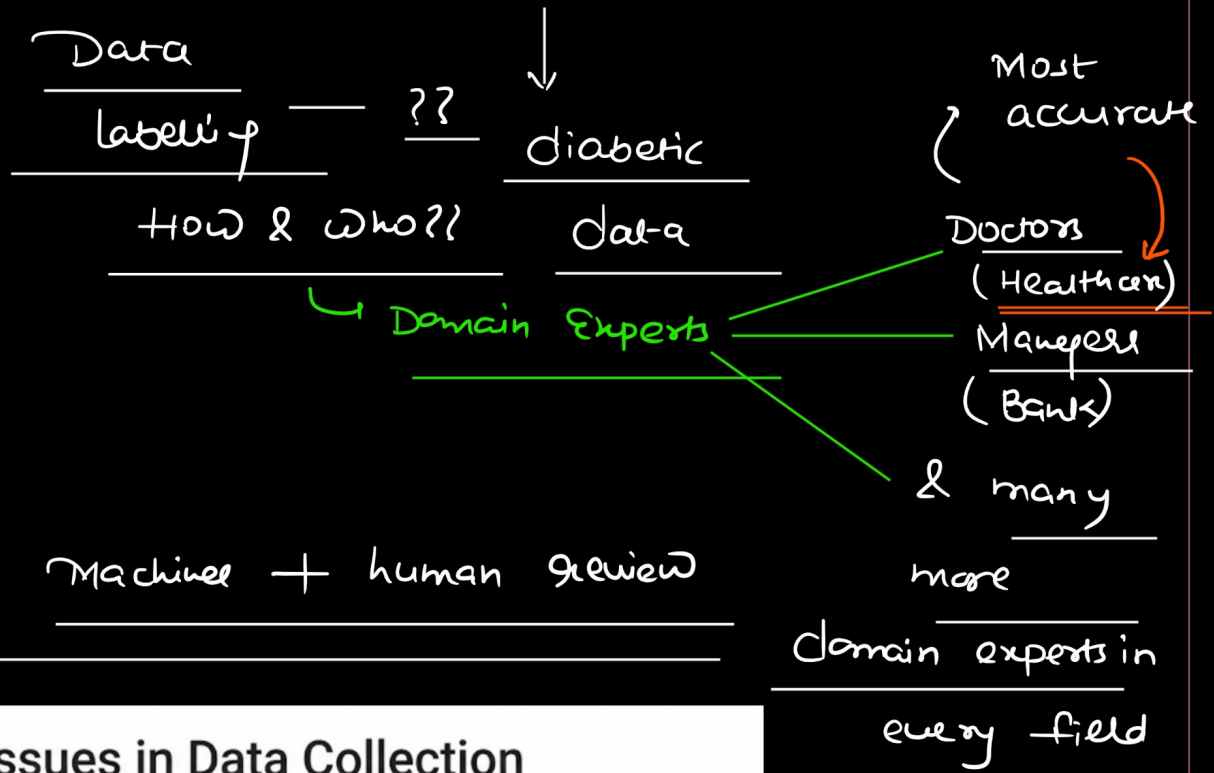


Machine Learning

supervised

unsupervised

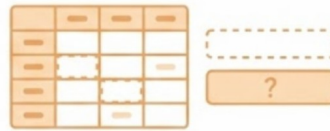
↳ labelled
data



Issues in Data Collection



1. Noise
Unwanted or irrelevant data



2. Missing Values
Incomplete data points



3. Unbalanced Data
Unequal category distribution



4. Bias
Data not fairly representative

Diabetic data

- ↳ Wrong entries
- ↳ missing values
- ↳ 0 → 900
- ↳ 1 → 100
- ↳ biases

Data Management & Cleaning



Data cleaning → forms, sensors, websites,
manual entry

typing, missing values,

dtype mismatch (date ← string) → DATE type

India/IND/Bharat
↳ India

Outliers

↳ extreme value

survey of salary earning in Bengaluru of
a data scientist

✓
5000, 50000, 70000, 85000, 10000, 1000000 → outlier

detect the outlier & handle it
in an effective way

Handle these

Problems

→ Handling of missing
Value

Missing % is very
Value

less → remove it completely

otherwise, we can impute the
missing value with a fixed
number

city name → Null value
↳ unknown

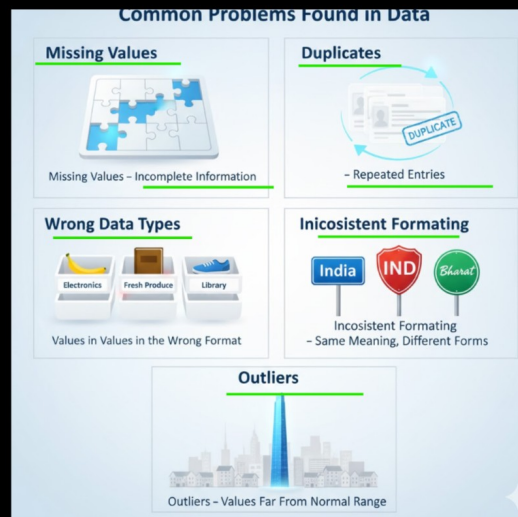
{ 20, 30, 45, 52, 63, 65, 77 }

numeric data → Statistics
Concepts

{ 20, 45, 30, 77, 52, 65, 63 } ← age of
a person (Average)
(Mean/Median)

$$(20 + 45 + 30 + 77 + 52 + 62 + 63) / 7$$

52



fraud

transaction

data

↳ regular → 20K to 70K

↳ fraud → 5 lakh (anomaly)

Outliers → errors (remove or handle)

Anomaly → important data point

↳ Deal With Duplicates

recreate some new column.

<u>customer ID</u>	<u>Date</u>

ignore / remove the
duplicate
records

↳ correcting the wrong data type

25 (text) → 25 (numeric)

↳ fix inconsistent formatting

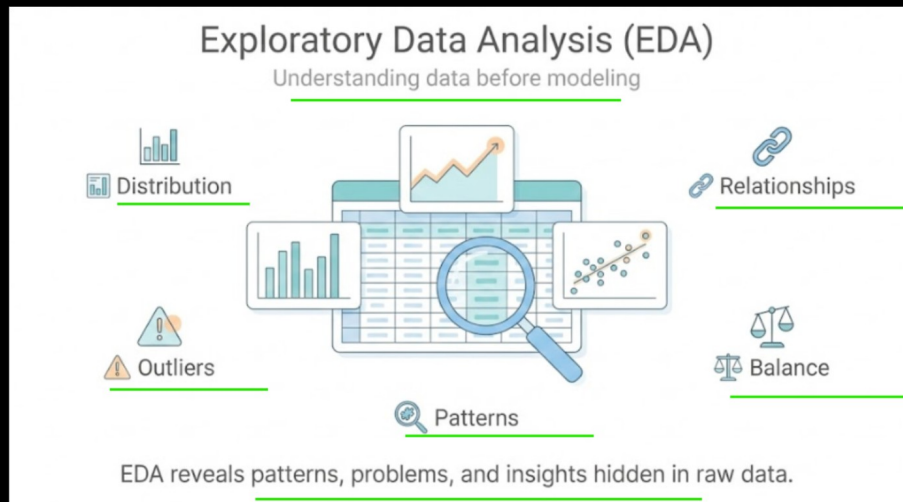
↳ standardize the text

↳ standardize the numeric
column into one specific
type

↳ Handle outliers → impute or drop the outliers if the % of outliers is very low

Data cleaning & EDA

EDA (Exploratory Data Analysis)



EDA helps us answer essential questions like:

1. What does my data look like?
2. Is it balanced or skewed?
3. Are there unexpected patterns or unusual values?
4. What relationships exist between different variables?
5. Does the data support the problem I want to solve?

Why EDA is essential
EDA is done because:

1. Raw numbers do not reveal everything. Patterns become visible only when we summarize or visualize the data.
 2. EDA tells us whether the dataset is trustworthy. Unexpected spikes or gaps may signal collection issues.
 3. EDA helps us discover valuable patterns. These patterns often guide business decisions or feature engineering steps.
 4. EDA helps identify relationships between variables. For example, higher age → more spending, or higher traffic → lower sales.
 5. EDA suggests which type of model might work best. If relationships appear linear, then linear models may work; if patterns are irregular, decision trees might fit better.
 6. EDA helps us catch errors early. It's better to detect issues now than after building a model.
- In short: EDA builds understanding, direction, and confidence.

Type of EDA

① Univariate Analysis → study one variable

- ↳ Distribution of data
(skewed / normal)
- ↳ Outliers
- ↳ balance within categories

② Bivariate Analysis → Relationship b/w

(Study two Variable together)

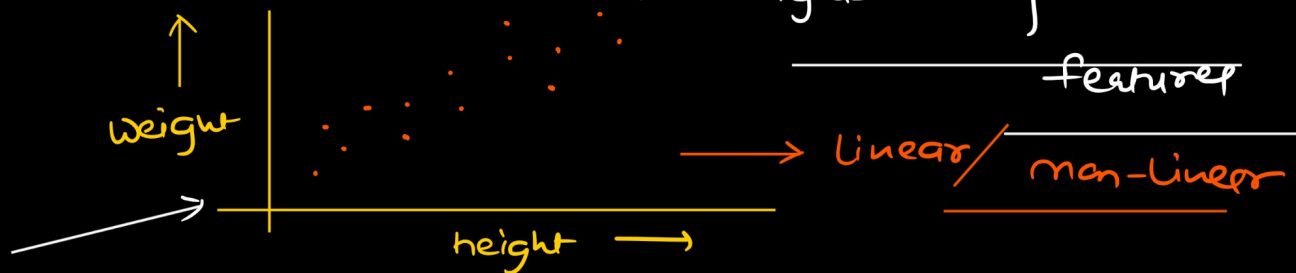
two variables

heatmap → -1 to $+1$
Correlation → > 0.85
income ↑ Spending ↑ (Positive Correlation)

scatterplot → to get the relationship

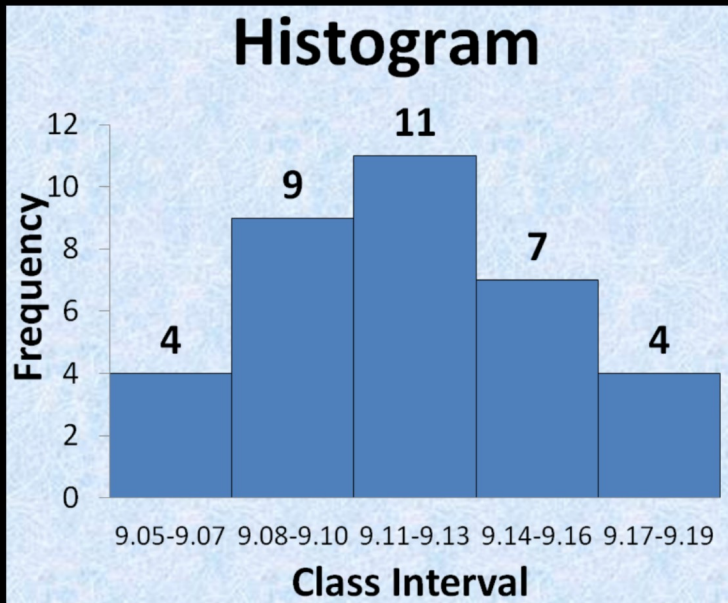
in height among two

feature

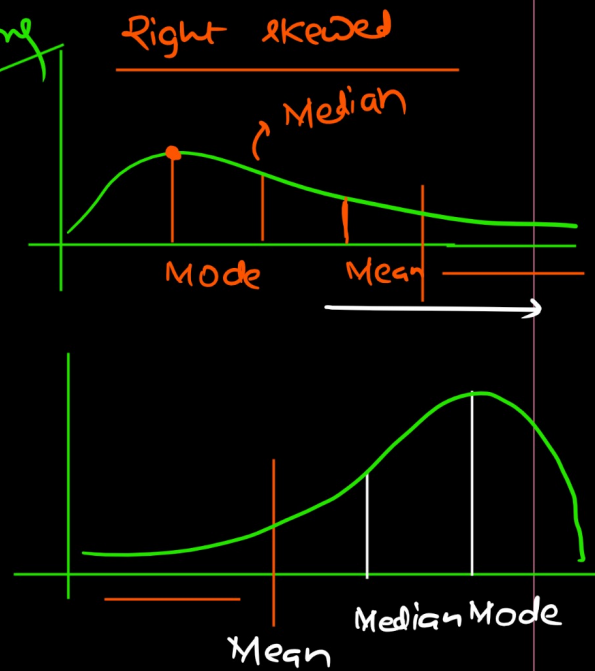
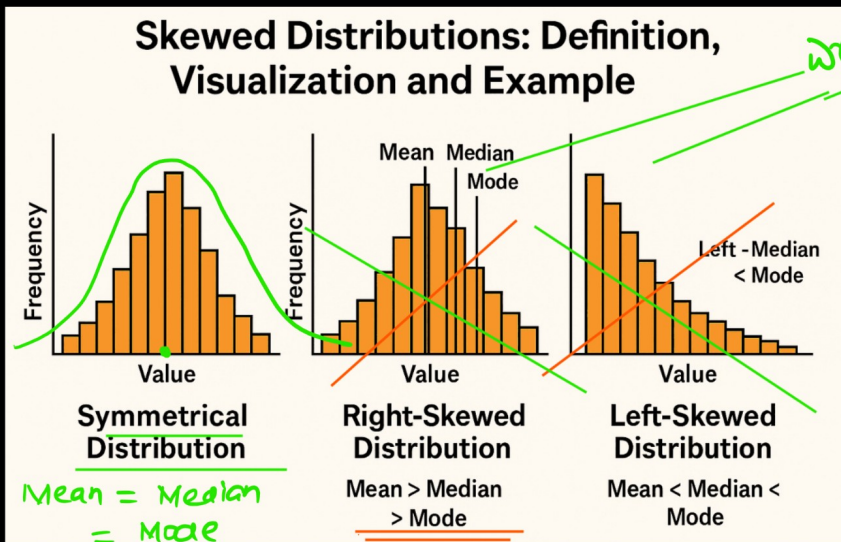


③ Multivariate Analysis → > 2 features

Common Visual



$$\text{Mode} < \text{Median} < \text{Mean}$$



$$\text{Mean} < \text{Median} < \text{Mode}$$

Feature Engineering

Feature engineering is the step where we shape the data so that a model can actually learn from it.

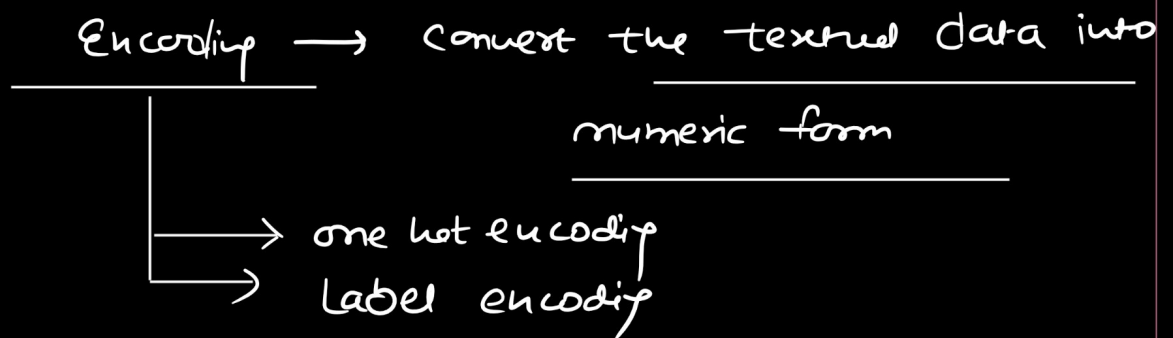
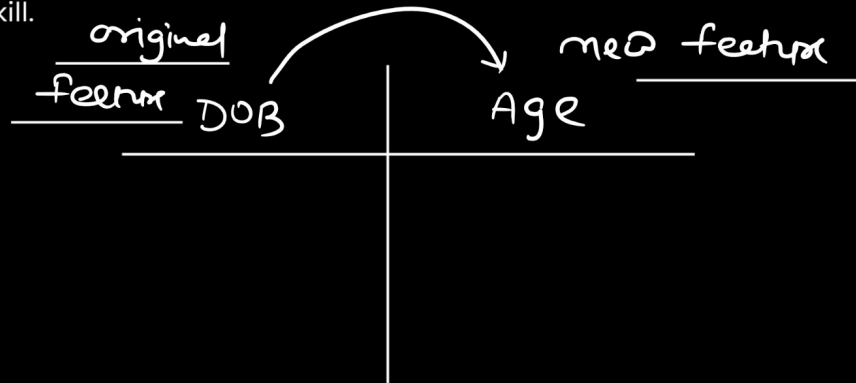
We do not just "feed the dataset" to a model and hope for the best. Instead, we carefully think about:

what information each column carries ←

whether that information is in a useful form ←

what new information we can create from existing columns ←

Feature engineering is often where human understanding of the problem is combined with data. That is why it is such an important skill.



GenAI / Textual data
↳ preprocessing

↳ tokens +

numeric

representation
of tokens